

Placing Storage Energies for Enhancing Small-Signal Stability of Converter-Based-Renewable Systems

Hui Yuan , Yi Hao , Huanhai Xin , Senior Member, IEEE, Linbin Huang , Member, IEEE, Yuhuan Zhou , Xiaofei Wang , Chunmeng Chen, Guoqiang Lu, Linan Qu, and Di Wu , Member, IEEE

Abstract—Storage energies (SEs) have the flexibility of bi-directional power regulation, so they have been integrated into the power system via power electronic converters to reduce the power fluctuations introduced by converter-based renewables (CBRs), such as solar and wind. However, converter-based SEs may interact with CBRs through the power network, which increases the complexity and difficulty of the phase-lock-loop (PLL)-induced small-signal stability analysis in multi-CBR systems, especially in weak grids. In this context, it remains unclear how the placement of SEs influences the PLL-induced small-signal stability, particularly when the SEs absorb active power from grids. To fill this gap, this paper analyzes the impact of SEs on the PLL-induced small-signal stability in multi-converter systems from the perspective of grid strength and proposes a method for optimally placing SEs to enhance the small-signal stability. First, the analytic results reveal that when the SEs absorb active power from the power grid, they can enhance grid strength and thus the PLL-induced small-signal stability; moreover, the degree of system stability improvement depends on the location of SE placement. On this foundation, a grid-strength-based method is proposed for the PLL-induced small-signal stability improvement via SE placements. The proposed method is validated based on three test systems. This paper provides an effective way of better understanding the interaction

among SEs and CBRs through power network, and coordinating the placements of SEs in future converter-dominated power systems.

Index Terms—Converter-based renewable, generalized short-circuit ratio (gSCR), small-signal stability, storage energy.

I. INTRODUCTION

WITH the development of converter-based renewables (CBRs) such as wind and solar, the modern power systems are shifting into converter-dominated systems (or multi-CBR systems) [1], [2]. However, the increasing penetration of CBRs is challenging the system's stability and security, e.g., power and energy balance issues caused by the fast fluctuation and unpredictable characteristics of renewables [3]. To address the issues of power and energy balance in the multi-CBR system, storage energies (SEs) are good choices, due to their high flexibility of bi-directional power regulation. Therefore, the large-scale applications of SEs in the multi-CBR system are recently achieving great attention [3], [4], [5], [6].

Recently, there have been some works on the impact of SEs on the security and stability of the CBR-dominated power systems [7], [8], [9]. For instance, Ref. [8] pointed out that the SEs can alleviate the low-voltage problems, but may aggravate transient overvoltage problems. Ref. [9] demonstrated that the SEs can effectively provide frequency regulation and dampen the low-frequency oscillation. Besides, there are also some works focused on the optimal placement and sizing of SEs [10], [11], [12] to improve the security and stability of CBR-dominated power systems. For instance, Ref. [10] proposed a method to minimize hourly cost and maximize the wind power utilization over a scheduling period. Ref. [12] focused on how to improve the system's transient stability by SE placements.

However, it has not been studied yet how the SEs (when absorbing active power) interact with CBRs and influence the small-signal stability, resulting from the interaction among fast CBR controls and power network in multi-CBR systems. The CBRs are commonly integrated into the power grid via power electronic converters with grid-following controls, which synchronizes with the grid through phase-lock loops (PLLs). These CBRs show good dynamic performance in stiff grids. However, the increasing renewable integration decreases grid strength and intensifies the interaction among fast CBR controls and power network. This may cause sub/sup-frequency oscillation issues or PLL-induced small-signal stability issues, especially in weak

Received 9 March 2024; revised 29 October 2024; accepted 6 January 2025. Date of publication 28 February 2025; date of current version 9 July 2025. Paper 2024-PSEC-0369.R1, presented at the 2023 IEEE 6th International Electrical and Energy Conference, Hefei, China, May 12–14, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Power Systems Engineering Committee of the IEEE Industry Applications Society [DOI: 10.1109/CIEEC58067.2023.10166980]. This work was supported in part by the National Natural Science Foundation of China under Grant 52207140 and Grant U24B6008, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LY23E070003, and in part by the National Science Foundation under Grant 2333563. (Corresponding author: Yi Hao.)

Hui Yuan, Huanhai Xin, and Linbin Huang are with the College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China, and also with the Zhejiang Key Laboratory of Electrical Technology and System on Renewable Energy, Hangzhou 310027, China (e-mail: yuan_hui@zju.edu.cn; xinh@zju.edu.cn; hlinbin@zju.edu.cn).

Yi Hao is with State Grid Shandong Electric Power Research Institute, Jinan 250002, China (e-mail: hy17860623037@163.com).

Yuhuan Zhou is with China Electric Power Planning Engineering Institute, Beijing 100032, China (e-mail: zhouyuhuan@zju.edu.cn).

Xiaofei Wang is with the College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: eewangxf@zju.edu.cn).

Chunmeng Chen and Guoqiang Lu are with Qinghai Electric Power Corporation, Xining 810001, China (e-mail: chenchunmeng@163.com; lgqiang2015@163.com).

Linan Qu is with the Science and Technology Research Institute of China Electrical Equipment Group, Shanghai 200040, China (e-mail: qu_ln@hotmail.com).

Di Wu is with the Department of Electrical and Computer Engineering, North Dakota State University, Fargo, ND 58102 USA (e-mail: di.wu.3@ndsu.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIA.2025.3546196>.

Digital Object Identifier 10.1109/TIA.2025.3546196

grids [13], [14], [15], [16]. The converter-based SEs are commonly integrated with the grid via power electronic converters with similar control configurations (i.e., grid-following control) to those of CBRs [7], [17]. That's to say, these SEs may interact with CBRs, which increases the complexity and difficulty of analyzing the PLL-induced small-signal stability issues in a multi-CBR system.

There have been many works about investigating the PLL-induced small-signal stability issues [13], [14], [15], [16], [18], [19], [20], [21]. For instance, Ref. [15] used the modal analysis to investigate how the short-circuit ratio (SCR) and PLL's parameters influence the small-signal stability. It demonstrated that the decrease of SCR and PLL's proportional parameter deteriorates the small-signal stability. Ref. [16] proposed an impedance-based method to reveal that how the CBR's power factor and current output influence the small-signal stability. In these works, the small-signal stability analysis is based on a single-CBR infinite-bus system. To analyze the impacts of the interaction among interconnected CBRs on the PLL-induced small-signal stability in a multi-CBR system, Ref. [18], [19], [20], [21] proposed a generalized SCR (gSCR)-based method from the viewpoint of grid strength.

However, these previous works do not consider the impact of SEs on the PLL-induced small-signal stability in a multi-CBR system. The main difference among SEs and CBRs is that SEs have the characteristic of bi-directional active power regulations, but CBRs only inject active power into the grid. When injecting active power into the grid, the SEs show similar dynamics as CBRs. In this case, it is known that the connection of SEs weakens grid strength (i.e., decreases SCR) and intensifies the interaction among PLL-based controls and networks [18], [19], [20], [21], which deteriorates the small-signal stability. When the SEs absorb active power from the grid, the dynamics of SEs are different from CBRs. Under this condition, the SEs may also interact with CBRs and thus influence PLL-induced small-signal stability, which has not been studied before.

To fill this gap, this paper investigates the impact of the interaction among SEs and CBRs on the PLL-induced small-signal stability, especially when SEs absorb active power. Moreover, we propose a method for effectively placing SEs to enhance the PLL-induced small-signal stability. More specifically, we model the multi-CBR system with SEs for the small-signal stability analysis. Then, based on modal decomposition theory, we prove that the small-signal stability of the multi-CBR system can be characterized by an equivalent single-CBR system, which avoids establishing the detailed system model. On this basis, we extend the gSCR-based method (our previous work referring to [18]) to be suitable for small-signal stability analysis of the multi-CBR system with SEs. The proposed method can consider the scenarios that the SEs inject (or absorb) active power into (or from) the grid. Based on the gSCR-based method, we explicitly show that the integration of SEs (when absorbing active power from the grid) is equivalent to enhancing grid strength (or improving the small-signal stability). Moreover, we explore how to optimally place SEs to enhance the small-signal stability of the multi-CBR system from the perspective of grid strength (i.e., gSCR).

The main contributions of this paper are summarized as:

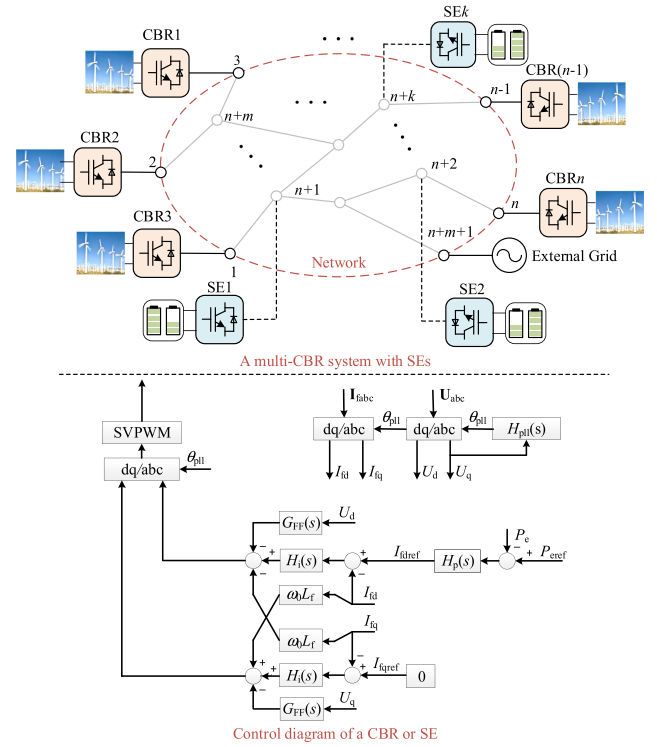


Fig. 1 Diagram of a multi-CBR system with SEs, and control of CBR (or SE).

- 1) We reveal the theoretical relationship between SEs and the PLL-induced small-signal stability in a multi-converter system from the perspective of grid strength. This provides a novel method for better understanding the interaction among SEs and CBRs through network in future converter-dominated power systems.
- 2) It is rigorously proven that when the SEs absorb active power from the power grid, they can enhance grid strength and thus system's small-signal stability, and the degree of system stability improvement depends on the location of SE placement.
- 3) A method is proposed for optimally placing SEs to enhance the system small-signal stability. The proposed method is suitable for a large power system integrated with many CBRs.

The rest of this paper is organized as follows. Section II introduces the concerned problems of this paper. Section III analyzes the system's small-signal stability. Section IV proposes a gSCR-based method to reveal the interaction among SEs and CBRs, and to analyze the interaction on the system's stability. Section V investigates the optimal placement of SEs. Section VI provides the simulation results. Conclusions are given in Section VII.

II. PROBLEM DESCRIPTION

Let us consider the multi-CBR system with k SEs as shown in Fig. 1, where nodes 1~ n represent CBR nodes, node $n+m+1$ is connected to external grids (simplified as the ideal voltage source), the remaining nodes are passive nodes. Since the actual SEs are not directly connected with CBRs, we assume that there are k SEs connected to passive nodes $n+1 \sim n+k$, ($k \leq m$). The

control configurations of CBRs and SEs are given in Fig. 1, where we mainly consider that the CBRs and SEs use the grid-following control synchronized with grids through PLL. Specifically, the SEs with grid-following control have similar control configurations as the CBRs. The difference is that the SEs have two operating forms: 1) inject active power into the grid; 2) absorb active power from the grid. When operating at form 1), the SEs can be considered as CBRs. Since the small-signal stability of the multi-CBR system has been investigated in previous works, we will omit the analysis of this operating form. When operating at form 2), how the SEs influence the system's small-signal stability has not been investigated, which is the one of focuses of this paper and will be analyzed in Section III. To simplify the analysis, we assume that: 1) the CBRs and SEs have the same control configurations and control parameters in the following, where the active power outer loop uses the constant active power loop (i.e., the difference between CBRs and SEs is the direction of active power output); 2) the converters' terminal voltage is nearly 1 p.u..

In this paper, we mainly focus on two issues for the small-signal stability of a multi-CBR system with SEs:

- 1) How do the SEs interact with CBRs and influence the system's small-signal stability, when the SEs operate at the scenario of absorbing active power from grids?
- 2) How to place the SEs for effectively enhancing the system's small-signal stability?

To deal with Issue 1), a grid-strength-based method will be proposed to investigate the small-signal stability of the multi-CBR system with SEs in Section III; besides, Section IV will propose a method to address Issue 2).

III. SMALL-SIGNAL STABILITY ANALYSIS OF MULTI-CBR SYSTEMS WITH SEs

In this section, we first derive the closed-loop characteristic equation of the multi-CBR with SEs for small-signal stability analysis. Then, by modal decomposition theory, an equivalent single-CBR system is proved to represent the small-signal stability of the original multi-CBR system with SEs.

A. Modelling of a Multi-CBR System With SEs

The multi-CBR system with SEs in Fig. 1 can be modeled in two parts: the power network, and the sources including CBRs and SEs. The network dynamics can be modeled as [21]

$$\Delta \mathbf{I}_{xy}(s) = \mathbf{B}_{\text{red}} \otimes \boldsymbol{\gamma}(s) \Delta \mathbf{U}_{xy}(s) \quad (1)$$

$$\boldsymbol{\gamma}(s) = \frac{\omega_0}{s^2 + \omega_0^2} \begin{bmatrix} s & \omega_0 \\ -\omega_0 & s \end{bmatrix} \quad (2)$$

where Δ denotes the perturbed value of a variable; $\mathbf{I}_{xy} \in \mathbb{R}^{2(n+k)}$ and $\mathbf{U}_{xy} \in \mathbb{R}^{2(n+k)}$ are vectors of node current injection and voltages in global xy-frame; $\mathbf{I}_{xy} = [\mathbf{I}_{1xy}, \mathbf{I}_{2xy}, \dots, \mathbf{I}_{(n+k)xy}]^T$, $\mathbf{U}_{xy} = [\mathbf{U}_{1xy}, \mathbf{U}_{2xy}, \dots, \mathbf{U}_{(n+k)xy}]^T$, where $\mathbf{I}_{ixy} = [I_{ix}, I_{iy}]$, $\mathbf{U}_{ixy} = [U_{ix}, U_{iy}]$; I_{ix} , I_{iy} are x-axis and y-axis components of the injected current at node i ; U_{ix} , U_{iy} are x-axis and y-axis components of terminal voltage at node i ; $\mathbf{B}_{\text{red}} \in \mathbb{R}^{(n+k) \times (n+k)}$ is node-reduced susceptance matrix, in which only CBRs and SEs nodes are retained; $\boldsymbol{\gamma}(s) \in \mathbb{R}^{2 \times 2}$

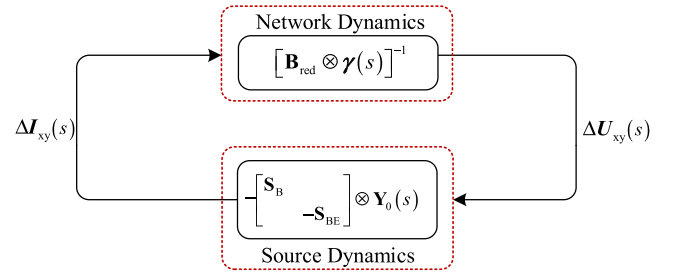


Fig. 2 Closed-loop dynamic model of a multi-CBR system with SEs.

represents dynamics of grid lines, and we assume that the grid line is purely inductive; $\otimes$ is the Kronecker product; ω_0 is rated synchronous frequency.

With the assumption that CBRs and SEs have the same control configurations and control parameters, the dynamics at the source side can be represented as (similar forms can refer to [18] and [21])

$$\Delta \mathbf{I}_{xy}(s) = - \begin{bmatrix} \mathbf{S}_B \otimes \mathbf{Y}_{\text{CBR}}(s) & \\ & \mathbf{S}_{\text{BE}} \otimes \mathbf{Y}_{\text{SE}}(s) \end{bmatrix} \Delta \mathbf{U}_{xy}(s) \quad (3)$$

where $\mathbf{S}_B = \text{diag}\{S_{B_i}\} \in \mathbb{R}^{n \times n}$, ($i = 1, \dots, n$), in which S_{B_i} is the rated capacity of CBR i ; $\mathbf{S}_{\text{BE}} = \text{diag}\{S_{\text{BE}_j}\} \in \mathbb{R}^{k \times k}$, ($j = 1, \dots, k$), in which S_{BE_j} is the rated capacity of SE j ; we note that this paper mainly focuses on the condition that CBRs output rated active power and SEs absorb rated active power from the grid; $\mathbf{Y}_{\text{CBR}}(s) \in \mathbb{R}^{2 \times 2}$ and $\mathbf{Y}_{\text{SE}}(s) \in \mathbb{R}^{2 \times 2}$ are the admittance matrix of each CBR and SE normalized at their own rated capacity, respectively.

According to [22] and [23], with the assumption that the terminal voltage is nearly 1 p.u., the impedance matrix of the CBR with grid-following control under varying active power output can be approximately written as

$$\mathbf{Y}_{\text{CBR}}(s) = P_e \mathbf{Y}_0(s) \quad (4)$$

where $\mathbf{Y}_0(s) \in \mathbb{R}^{2 \times 2}$ is impedance matrix of a CBR with rated active power output, P_e is CBR's active power output.

We can see from (4) that when the converter absorbs active power from the grid, the corresponding converter's impedance matrix only changes P_e , which becomes a negative real value. Therefore, the impedance matrix $\mathbf{Y}_{\text{SE}}(s)$ of SE j (when absorbing active power from the grid) can be rewritten as

$$\mathbf{Y}_{\text{SE}}(s) = -S_{\text{BE}} \mathbf{Y}_0(s) \quad (5)$$

Substituting (4) and (5) into (3), the dynamics at the source side can be rewritten as:

$$\Delta \mathbf{I}_{xy}(s) = - \left\{ \begin{bmatrix} \mathbf{S}_B & \\ & -\mathbf{S}_{\text{BE}} \end{bmatrix} \otimes \mathbf{Y}_0(s) \right\} \Delta \mathbf{U}_{xy}(s) \quad (6)$$

Combining (1) and (6), the close-loop dynamic model of the multi-CBR system with SEs can be given as Fig. 2. Therefore, the close-loop characteristic equation of the multi-CBR system with SEs can be given as

$$\det \left[\begin{bmatrix} \mathbf{S}_B & \\ & -\mathbf{S}_{\text{BE}} \end{bmatrix} \otimes \mathbf{Y}_0(s) + \mathbf{B}_{\text{red}} \otimes \boldsymbol{\gamma}(s) \right] = 0 \quad (7)$$

By left-multiplying $\mathbf{S}_{B1}^{-1} \otimes \mathbf{D}_2$ and right-multiplying $\mathbf{D}_{2(n+k)} \otimes \gamma^{-1}(s)$ (where $\mathbf{D}_i \in \mathbb{R}^{i \times i}$ represents an $i \times i$ identity matrix), (7) can be rewritten as:

$$\det(\mathbf{C}(s)) = \det[\mathbf{D}_n \otimes (\mathbf{Y}_0(s) \gamma^{-1}(s)) + (\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}) \otimes \mathbf{D}_2] = 0 \quad (8)$$

$$\mathbf{S}_{B1} = \begin{bmatrix} \mathbf{S}_B & \\ & -\mathbf{S}_{BE} \end{bmatrix} \quad (9)$$

where $\mathbf{C}(s) \in \mathbb{R}^{2(n+k) \times 2(n+k)}$, $\mathbf{S}_{B1} \in \mathbb{R}^{(n+k) \times (n+k)}$.

The obtained closed-loop characteristic (8) can be used to analyze the small-signal stability of the multi-CBR system with SEs.

B. Small-Signal Stability Analysis of the Multi-CBR System With SEs

It can be seen from Fig. 2 that it is difficult to analyze the small-signal stability by directly solving the characteristic (8), due to the high-dimension dynamic model. To address this issue, we will decompose the system's characteristic (8) into a set of equivalent single-converter systems, and use one of these decomposed single-converter systems to represent the small-signal stability of the multi-CBR system with SEs in the following.

By left-multiplying $\mathbf{T} \otimes \mathbf{D}_2$ and right-multiplying $\mathbf{T}^{-1} \otimes \mathbf{D}_2$ (where $\mathbf{T}(\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}) \mathbf{T}^{-1} = \mathbf{A}$, $\mathbf{T} \in \mathbb{R}^{(n+k) \times (n+k)}$, $\mathbf{A} = \text{diag}(\lambda_i)$, $i = 1, \dots, (n+k)$), (8) is decomposed as [18]:

$$\det(\mathbf{C}(s)) = \det(\mathbf{c}_1(s)) \cdots \det(\mathbf{c}_{n+k}(s)) = 0 \quad (10)$$

$$\det(\mathbf{c}_i(s)) = \det[\mathbf{Y}_0(s) \gamma^{-1}(s) + \lambda_i \mathbf{D}_2] = 0, \lambda_i = 1, \dots, n+k \quad (11)$$

where λ_i is the eigenvalue of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$, $\det(\mathbf{c}_i(s)) = 0$ is the close-loop characteristic equation of a single-converter system with the SCR equal to λ_i , $\mathbf{c}_i(s) \in \mathbb{R}^{2 \times 2}$.

We can see from (10) that the multi-CBR system with SEs can be decomposed as $n+k$ single-converter systems for small-signal stability analysis. These n single-converter systems have the same converter dynamics, but different SCRs. Therefore, the small-signal stability of the multi-CBR system with SEs can be represented by one of these single-converter systems. Before introducing the corresponding single-converter system, a **Proposition** about eigenvalues of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$ is given as follows:

Proposition 1: The matrix $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$ satisfies that:

- 1) The eigenvalues of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$ are all real values;
- 2) it has n positive eigenvalues and k negative eigenvalues.

Proof: The detailed proof refers to Appendix I-A. ■

According to Proposition 1, it is concluded that in these decoupled single-converter systems in (10), there are n systems with positive SCRs, but the others have negative SCRs. By experiments, we find that when the SCR is negative (i.e., the converter absorbs active power from the grid) in a single-converter system, the system is stable in weak grids. The detailed discussion will be given in Section V. In other words, the small-signal stability of the multi-CBR system with SEs can be represented by the single-converter system with the smallest positive eigenvalue of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$ in these decoupled single-converter systems in (10), which is also a single-CBR system.

IV. GSCR-BASED METHOD FOR ANALYZING THE IMPACT OF SEs ON SYSTEM'S SMALL-SIGNAL STABILITY

In this section, we first propose a gSCR-based method for small-signal stability margin assessment of a multi-CBR system with SEs from the viewpoint of grid strength. Then, an implementation process of the proposed method is provided.

A. gSCR-Based Method for System's Stability Assessment

As discussed in Section III, the small-signal stability of the multi-CBR system with SEs can be represented by a single-CBR system with the smallest positive eigenvalue of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$, denoted as λ_{min}^+ . When the control configuration and control parameters of CBRs and SEs are given, the system's small-signal stability can be determined by λ_{min}^+ . That is, the larger λ_{min}^+ is, the system is more stable; otherwise, the decrease of λ_{min}^+ deteriorates the system's small-signal stability. Therefore, we define λ_{min}^+ as gSCR to evaluate the small-signal stability of the multi-CBR system with SEs in the following:

Definition 1: The gSCR is defined as the minimal positive eigenvalue of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$, given as

$$\text{gSCR} = \lambda_{min}^+ \{ \mathbf{S}_{B1}^{-1} \mathbf{B}_{red} \} \quad (12)$$

where $\lambda_{min}^+ \{ \}$ denotes minimal positive eigenvalue of a matrix.

Besides, we define the critical gSCR (named as CgSCR) to represent the system is critically stable in the following:

Definition 2: The CgSCR is defined as the value of gSCR when the system is critically stable, given as

$$\text{CgSCR} = \arg_{\text{gSCR}} \{ \det(\mathbf{Y}_0(s_c) + \text{gSCR} \cdot \mathbf{D}_2) = 0 \} \quad (13)$$

where $\arg \{ \}$ represents eigenvalue calculation for CgSCR; $s_c = j\omega_c$ is the system weakest eigenvalue, located on the imaginary axis; ω_c is the critical oscillation frequency in rad/s.

Based on gSCR and CgSCR, the stability of the multi-CBR system with SEs is evaluated as: if $(\text{gSCR} - \text{CgSCR}) = 0$, the system is critically stable; if $(\text{gSCR} - \text{CgSCR}) > 0$, the system is stable; the larger the $(\text{gSCR} - \text{CgSCR})$ is, the system is more stable; otherwise, if $(\text{gSCR} - \text{CgSCR}) < 0$, the system is unstable.

Moreover, we can conclude from (12) and (13) that the SEs mainly influence the gSCR, but has no impact on the CgSCR. Proposition 2 is given to demonstrate whether the SEs increase or decrease the gSCR in the following:

Proposition 2: The connection of SEs will increase gSCR, when the SEs operate at the scenario of absorbing active power.

Proof: The detailed proof refers to Appendix I-B. ■

Proposition 2 demonstrates that the SEs when absorbing active power from the grid will improve the system's small-signal stability (i.e., increase gSCR). Besides, when the SEs are located at different nodes, the updated gSCR is different. In other words, the location of SEs determines how much the SEs improve the stability, which will be discussed in Section V.

B. Implementation of gSCR-Based Method

A detailed process is given to describe how to evaluate the small-signal stability margin of the multi-CBR system with SEs from the viewpoint of grid strength as shown in Fig. 3:

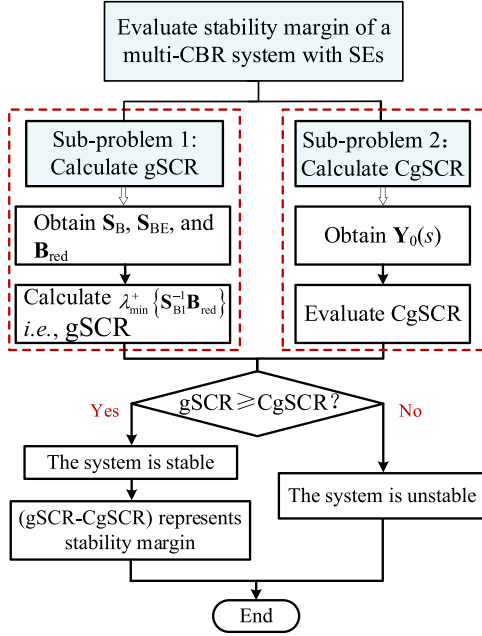


Fig. 3. Process for evaluating small-signal stability margin of a multi-CBR system with SEs.

- 1) We firstly decompose the complex small-signal stability margin evaluation issue into two simple sub-problems: 1) how to calculate gSCR; 2) how to calculate CgSCR.
- 2) Solve sub-problems 1) and 2). For sub-problem 1), we firstly obtain the capacity matrices $\mathbf{S}_B$ and $\mathbf{S}_{BE}$, and network admittance matrix $\mathbf{B}_{red}$. Then, we use $\mathbf{S}_B$, $\mathbf{S}_{BE}$ and $\mathbf{B}_{red}$ to calculate gSCR by (12). For sub-problem 2), we consider two scenarios: a) the control parameters of CBRs and SEs are known; b) the control parameters of CBRs and SEs are unknown. In scenario a), we can directly calculate CgSCR by (13). In scenario b), we can evaluate CgSCR by experiment. To be specific, we firstly establish a single-CBR system with the given “black-box” model. Then, we change the SCR of the established single-CBR system until the system is critically stable by observing time-domain responses. The corresponding value of the SCR is CgSCR.
- 3) Evaluate the small-signal stability margin by (gSCR - CgSCR). If (gSCR - CgSCR) = 0, it demonstrates that the system is critically stable; if (gSCR - CgSCR) < 0, it demonstrates that the system is unstable; if (gSCR - CgSCR) > 0, it demonstrates that the system is stable; besides, the larger (gSCR - CgSCR) is, the system is more stable.

V. PLACING SEs TO ENHANCE SYSTEM'S STABILITY

A. Optimization Problem for Placing SEs and Its Algorithm

We investigate how to optimally place SEs to enhance the system's small-signal stability margin. It is known that the active power outputs of CBRs are fluctuant, due to the climate change. When only a few CBRs output active powers (named as low operating mode), the CBRs' active power outputs may not satisfy the demand of the grid. In this case, the SEs are required to inject active powers into the grid for filling the gap of the grid's

demand. When all CBRs output nearly the rated active powers (named as large operating mode), the CBRs' active power outputs may exceed the grid's demand. In this case, the SEs are required to absorb the excess power generation of CBRs. For the low operating mode, the grid strength is commonly stiff and the system is small-signal stable. However, for the large operating mode, the grid strength is weak and the system may suffer the PLL-induced small-signal instability issues. Therefore, in the following, we mainly investigate how to optimally place SEs to enhance the system's small-signal stability margin under the maximal operating mode, i.e., all CBRs output the rated active powers and the SEs absorb the rated active powers.

More specifically, we consider a power system with n CBRs and k SEs will be installed in the system, where the capacity of each SE is assumed to be equal and k is set to satisfy the grid's demand from the perspective of active power balance. The optimal locations to place these SEs can be obtained by solving the following optimization problem:

$$\min_{I \subset V} \lambda_{\max}^+ \{ \mathbf{M}_{V(I)} \} \quad (14)$$

where $V = \{n+1, \dots, n+m\}$ is the set of passive nodes; $I \subset V$ is the set of k nodes to place SEs which will be solved by optimization problem (14); $\mathbf{M} = \mathbf{S}'_{B1} \mathbf{Z} \in \mathbb{R}^{(n+m) \times (n+m)}$, which consider the connection of k SEs. Since $\text{gSCR} = 1/\lambda_{\max}^+ \{ \mathbf{M}_{V(I)} \}$, (according to (19)), (14) is aimed at selecting the location of SEs to maximally increase the gSCR and thus the system's small-signal stability.

Equation (14) is a combinatorial optimization problem, which may suffer huge computational burden for a power system with large-scale CBRs. To address this issue, we propose a heuristic but simple method to obtain a sub-optimal solution. More specifically, we replace the optimization problem (14) with the following iterative optimization problems that will be solved for k times:

$$\min_{\alpha_j \in V} \lambda_{\max}^+ \{ \mathbf{M}_{V(I)}^{[j]} \} \quad (15)$$

where $j \in \{1, \dots, k\}$ denotes the j th iteration, $\alpha_j \in V$ is the node to place the SE that will be determined by the j th iterative optimization problem (15), $\mathbf{M}_{V(I)}^{[j]}$ is a function to consider the connection of j th SE in $\mathbf{M}$, wherein $\mathbf{M}^{[j+1]} = \mathbf{M}_{V(I)}^{[j]}$ with $\mathbf{M}^{[1]} = \mathbf{M}$. By iteratively solving the optimization problem (15) for k times using the heuristic method, we can obtain the locations for placing k SEs.

The iterative optimization problem in (14) can be solved by calculating the sensitivity of the maximal eigenvalue of $\mathbf{S}'_{B1} \mathbf{Z}$ for SE j 's capacity and choosing the minimal one to place the SE. The reason is that the sensitivity of $\mathbf{S}'_{B1} \mathbf{Z}$'s maximal eigenvalue for SE j 's capacity can be written as

$$\frac{\partial \lambda_{\max}^+}{\partial S_{BEj}} = -a \lambda_{\max}^+ v_{j1}^2 \leq 0 \quad (16)$$

where $\lambda_{\max}^+ = \lambda_{\max}^+ \{ \mathbf{M}_{V(I)}^{[j-1]} \}$; v_{j1} is the j th element of $\mathbf{v}_1$, wherein $\mathbf{v}_1$ is the left-eigenvector of $\lambda_{\max}^+$; a is a positive real number; the derivation of (16) refers to Appendix I-C.

We can see from (16) that the sensitivity of $\lambda_{\max}^+$ for S_{BEj} is non-positive. In other words, the connection of a SE always

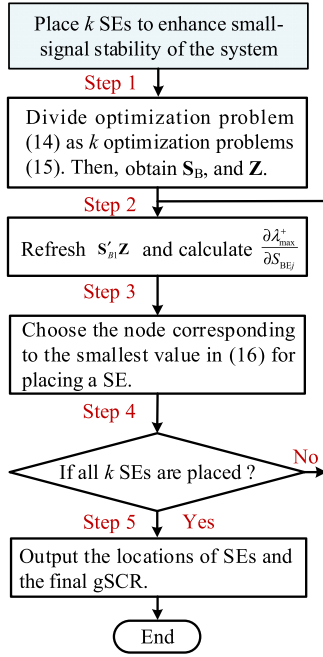


Fig. 4 Solving process for the optimization problem in (14).

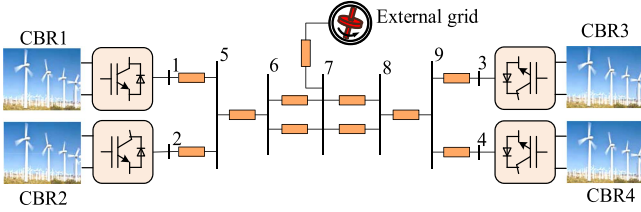


Fig. 5 A two-area four-CBR test system.

decreases $\lambda_{\max}^+$ (i.e., increases gSCR). Moreover, if $-a\lambda_{\max}^+ v_{j1}^2$ is smaller, $\lambda_{\max}^+$ decreases more due to the connection of SE_j , when S_{BEj} is given. Therefore, the optimal solution of (15) is that the SE is connected at the node with minimal $-a\lambda_{\max}^+ v_{j1}^2$.

The solving process for the optimization problem (14) as shown in Fig. 4 can be summarized as follows:

- 1) The optimization problem (14) is decomposed as k iterative optimization problems (15). Then, we obtain the capacity matrix S_B for n CBRs and the network impedance matrix Z , where nodes $1 \sim n+m$ are remained;
- 2) Refresh the weighted network Laplacian matrix $S'_{B1}Z$ and calculate $\partial\lambda_{\max}^+/\partial S_{BEj}$, $j = n+1, \dots, n+m$, referring to (16);
- 3) Rank the value of $\partial\lambda_{\max}^+/\partial S_{BEj}$ from largest to smallest. Then, choose the node corresponding to the smallest value for placing a SE.
- 4) If all k SEs are placed in the multi-CBR system, go to step 5); otherwise, go to step 2).
- 5) Output the locations for k SEs and the final gSCR.

B. Illustrative Case

To illustrate the proposed method described in Fig. 4, let us consider a two-area four-CBR test system as shown in Fig. 5. In the following studies, the system's parameters are

TABLE I
SENSITIVITY OF $\lambda_{\max}^+$ FOR S_{BEj} ON EACH NODE IN FIRST ITERATION

Node	Sen	Node	Sen	Node	Sen
1	-0.175	4	-0.149	7	-0.008
2	-0.113	5	-0.061	8	-0.013
3	-0.067	6	-0.017	9	-0.031

TABLE II
SENSITIVITY OF $\lambda_{\max}^+$ FOR S_{BEj} ON EACH NODE IN SECOND ITERATION

Node	Sen	Node	Sen	Node	Sen
1	-0.024	4	-0.272	7	-0.007
2	-0.048	5	-0.024	8	-0.016
3	-0.104	6	-0.011	9	-0.044

TABLE III
gSCR (OR $1/\lambda_{\max}^+$) FOR DIFFERENT PLACEMENTS OF TWO SEs

Node	gSCR	Node	gSCR	Node	gSCR	Node	gSCR
(1,1)	3.01	(2,5)	3.02	(4,4)	2.91	(6,7)	2.75
(1,2)	3.06	(2,6)	2.98	(4,5)	3.21	(6,8)	2.77
(1,3)	3.36	(2,7)	2.96	(4,6)	2.95	(6,9)	2.83
(1,4)	3.57	(2,8)	2.99	(4,7)	2.90	(7,7)	2.71
(1,5)	3.03	(2,9)	3.11	(4,8)	2.91	(7,8)	2.74
(1,6)	2.99	(3,3)	2.93	(4,9)	2.93	(7,9)	2.79
(1,7)	2.98	(3,4)	2.98	(5,5)	2.97	(8,8)	2.75
(1,8)	3.01	(3,5)	3.13	(5,6)	2.90	(8,9)	2.87
(1,9)	3.14	(3,6)	2.91	(5,7)	2.88	(9,9)	2.85
(2,2)	3.03	(3,7)	2.87	(5,8)	2.90		
(2,3)	3.31	(3,8)	2.88	(5,9)	3.00		
(2,4)	3.47	(3,9)	2.91	(6,6)	2.78		

all in per-unit values. The control configurations of SEs and CBRs refer to Fig. 1 and control parameters are presented in Table IX in Appendix II, which refers to Refs. [18] and [20]. The CBRs' capacities and network parameters are given in Table X in Appendix II. The external grid is simplified as an infinite bus (i.e., an ideal voltage source). We consider the case that two SEs (each SE's rated capacity is 0.5 p.u.) will be placed in this system. Note that in the following, we focus on the operating conditions that the CBRs output (or the SEs absorb) rated active power into (from) the grid. Then, we will use this case to detail how to use the proposed method in Fig. 4 for placing SEs.

In step 1), the optimization problem (14) is decomposed as two iterative optimization problems (15).

In step 2), we first consider the first iterative optimization problem. The obtained $S'_{B1}Z$ is given in (17). The resulting $\lambda_{\max}^+$ and $\partial\lambda_{\max}^+/\partial S_{BEj}$ are given in (18) and Table I.

In step 3), we can see from Table I that node 1 is the optimal node for placing first SE. Then, we repeat steps 2) and 3) to solve the second iterative optimization problem. We refresh $S'_{B1}Z$ with the consideration of placing first SE. The resulting $\partial\lambda_{\max}^+/\partial S_{BEj}$ is given in Table II.

Table II shows that node 4 is the optimal node for placing second SE. Therefore, the obtained optimal placement of SEs are nodes 1 and 4.

To demonstrate the validity of the proposed method, we provide the values of $\lambda_{\max}^+$ for all possible conditions, when placing two SEs. The results are shown in Table III. We can see from Table III that the global optimal solution for placing

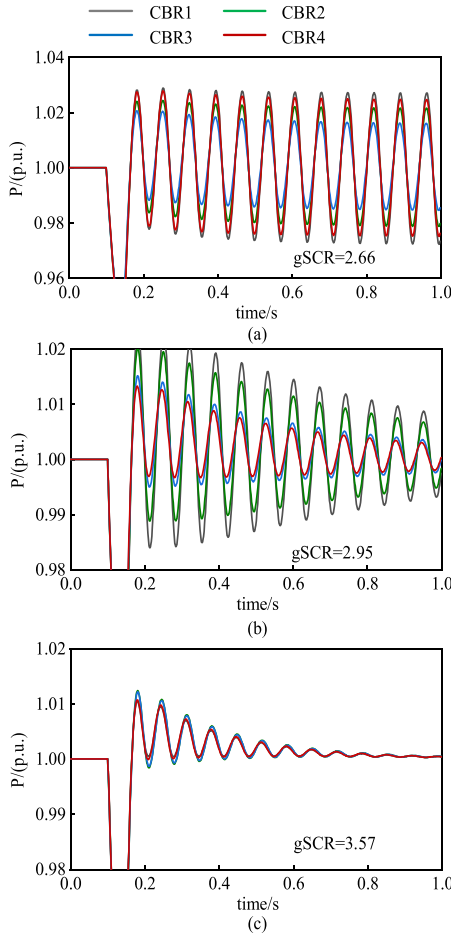


Fig. 6. Time-domain responses of active power outputs of all CBRs in the 9-node system under: (a) Case 1; (b) Case 2; (c) Case 3.

SEs are nodes 1 and 4, which is consistent with the results obtained by the proposed method. We will explore to calculate the rigorously optimal and computationally efficient solutions of the optimization problem (14) in the future work.

In the following, the time-domain simulation results are provided to further verify the validity of the optimal placement of SEs. Three cases are considered:

- 1) Do not consider the placement of SEs as a reference;
- 2) Place SEs at nodes 4 and 6;
- 3) Place SEs at nodes 1 and 4 by the proposed method.

Under these cases, the same disturbance is applied at the infinite bus at $t = 0.1$ s to cause 10% voltage drop and then is cleared at 0.02s later. Fig. 6 shows the time-domain responses of all CBRs' active power outputs under these cases. We can see from Fig. 6 that when applying two SEs, the oscillation curves of all CBRs become from undamped to convergent. This demonstrates that the placement of SEs can improve the system's small-signal stability. Moreover, when placing SEs at nodes 1 and 4, the system converges much faster, in comparison with the case that SEs are placed at nodes 4 and 6. This illustrates that the system is more stable when placing SEs by our proposed method, which verifies the superiority of our proposed method.

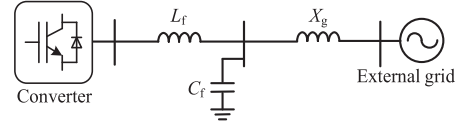


Fig. 7. Diagram of a single-converter infinite-bus system.

$$\mathbf{S}'_{B1}\mathbf{Z} =$$

$$\begin{bmatrix} 0.223 & 0.069 & 0.015 & 0.015 & 0.069 & 0.030 & 0.015 & 0.015 & 0.015 \\ 0.139 & 0.238 & 0.030 & 0.030 & 0.139 & 0.060 & 0.030 & 0.030 & 0.030 \\ 0.045 & 0.045 & 0.249 & 0.130 & 0.045 & 0.045 & 0.045 & 0.071 & 0.130 \\ 0.015 & 0.015 & 0.043 & 0.246 & 0.015 & 0.015 & 0.015 & 0.024 & 0.043 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (17)$$

$$\lambda_{\max}^+ = 0.376 \quad (18)$$

VI. SIMULATION RESULTS

In this section, we first provide a single-converter infinite-bus system to demonstrate that the system is stable under weak grids, when the converter absorbs active power from the grid. Then, a modified 39-node system are provided to verify the effectiveness of the proposed gSCR-based method and SE's optimal placement method. Finally, a 33-converter system is provided to verify the validity of the proposed SE's optimal placement method.

A. Case Studies on a Single-Converter Infinite-Bus System

We provide the modal analysis of a single-converter infinite-bus system in Fig. 7 to verify that the system is stable under weak grids, when the converter absorbs the active power from the grid. In Fig. 7, the external grid is simplified as an infinite-bus node, represented by an ideal voltage source; L_f and C_f are filter inductance and capacitor; X_g is the grid line's reactance; the converter's control configuration refers to Fig. 1 and corresponding control parameters refer to Table IX in Appendix II. Converter's two operating forms are considered, including that the converter outputs (or absorbs) the active power into (or from) the grid.

Fig. 8 shows the loci of system's weakest eigenvalues (i.e., the pair of eigenvalues that is nearest the imaginary axis) with SCR changing from -1.5 to -4.0 p.u. (or from 4.0 to 1.5 p.u.). Note that when the SCR is smaller than zero, it means that the converter absorbs active power from the grid. It can be observed from Fig. 8 that when the converter injects active power into the grid, the system's weakest eigenvalues move from the left-hand to the right-hand of the complex plane with the decrease of SCR, which is consistent with the conclusion in previous works [15], [18]. In comparison, when the converter absorbs active power from the grid, the system's weakest eigenvalues remain in the left-hand of the complex plane (far away from the imaginary

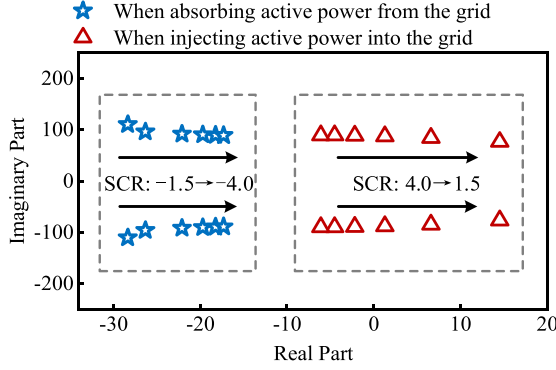


Fig. 8. Loci of weakest eigenvalues of a single-converter infinite-bus system with varying SCR.

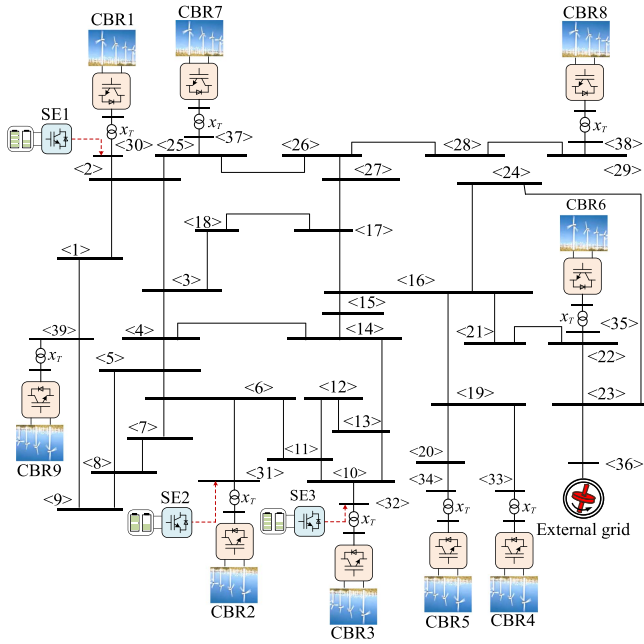


Fig. 9. Diagram of a modified 39-node system with 9 CBRs and 3 SEs.

axis) with varying SCR. The simulation results demonstrate that the single-converter system is strongly small-signal stable, when the converter absorbs active power from the grid. Therefore, in the decoupled $n+k$ single-converter systems in (10), the small-signal stability of the multi-CBR system with SEs is represented by the single-converter system with minimal positive eigenvalue of $\mathbf{S}_{B1}^{-1} \mathbf{B}_{red}$, i.e., gSCR.

B. Case Studies on a Modified IEEE 39-Node System

A modified IEEE 39-node system is provided as shown in Fig. 9. In this 39-node system, nodes 30~35, 37~39 are connected to CBRs through a set-up transformer, and node 36 is connected to the external grid simplified as an ideal voltage source. Three SEs are intended to be employed in this system. The network topology of the system is the same as that of the standard IEEE 39-node system, and network parameters are given in Table XI in Appendix II. The control parameters of all CBRs and SEs are presented in Table IX in Appendix II. The

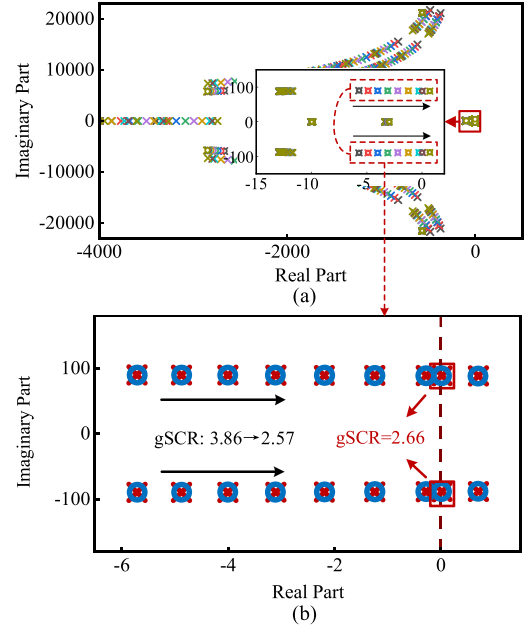


Fig. 10. Eigenvalue loci of the 39-node system and the decoupled single-CBR system with gSCR when increasing coefficient m : (a) loci of the system's all eigenvalues; (b) loci of system's weakest eigenvalues.

capacities of all CBRs are set as 1.0 p.u. and the capacities of all SEs are set as 0.75 p.u..

1) Verification of Proposed gSCR-Based Method:

a) *Modal analysis*: The proposed gSCR-based method for the small-signal stability analysis of the multi-CBR system with SEs is validated by modal analysis on the 39-node system, where each of nodes 30~32 is installed with a SE as shown in Fig. 9. Under this scenario, several cases are created by equally proportionally changing network lines' reactance with coefficient m , which is proportional to the line length in the system. When increasing coefficient m from 0.70 to 1.05 proportionally, we evaluate the value of gSCR, all eigenvalues and weakest eigenvalues of the 39-node system and decoupled single-CBR system with gSCR as shown in Fig. 10.

The results of Fig. 10 validate that the small-signal stability and stability margin of the multi-CBR system with SEs can be evaluated by the gSCR and its critical value, i.e., C_{gSCR} . According to Fig. 10(b), we can obtain that the C_{gSCR} is 2.66. Besides, when $gSCR = C_{gSCR}$, the weakest eigenvalues in the 39-node system are exactly located at the imaginary axis, which indicates that the system is critically stable. When $gSCR > C_{gSCR}$, the weakest eigenvalues are located on the left-hand of the complex plane, which suggests that the system is stable. Moreover, the larger $(gSCR - C_{gSCR})$ is, the system is more stable. When $gSCR < C_{gSCR}$, the weakest eigenvalues are located on the right-hand of the complex plane, which indicates the system is unstable. That is, the small-signal stability margin of the multi-CBR system with SEs can be identified by $(gSCR - C_{gSCR})$. Thus, the efficacy of the proposed gSCR-based method is verified.

Also, it can be observed from Fig. 10 that the small-signal stability of the 39-node system can be characterized by the

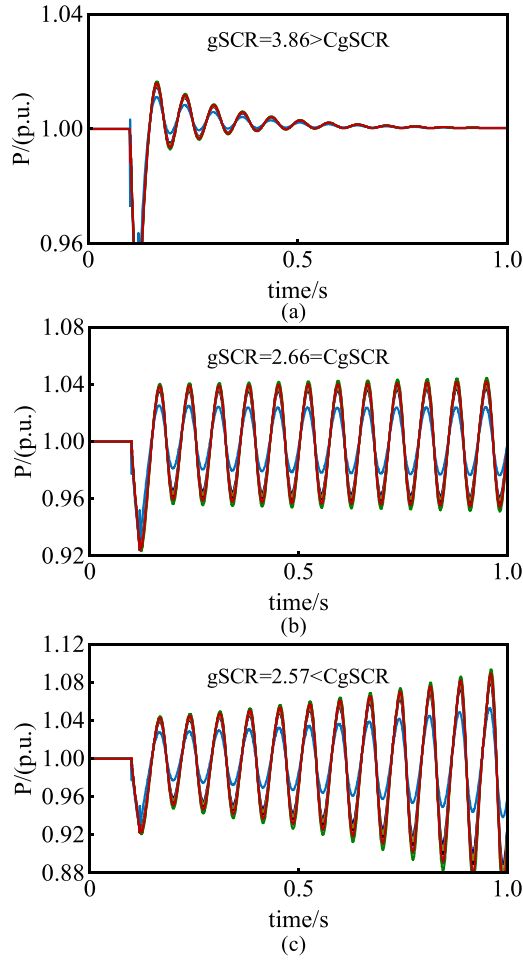


Fig. 11. Time-domain responses of active power outputs of all CBRs in the 39-node system when $gSCR = 3.86, 2.66, 2.57$.

decoupled single-CBR system with $gSCR$. As shown in Fig. 10(b), the weakest eigenvalues of the 39-node system with SEs are almost the same as those of the decoupled single-CBR system with $gSCR$. The observations agree with the analysis results in Section IV.

b) Time-domain simulation: In the 39-node system, the proposed $gSCR$ -based method for small-signal stability analysis of the multi-CBR system with SEs is further validated by the electromagnetic time-domain simulation. When coefficient m is set as 0.700, 1.015 and 1.050, we evaluate $gSCR$ are 3.856, 2.659 and 2.571, respectively. Under these conditions, a same disturbance is applied to the infinite bus, which causes the voltage drop by 10% at 0.10s and then is cleared at 0.12s. Fig. 11 shows the corresponding active power trajectories of all CBRs in the 39-node system under these conditions.

It can be observed from Fig. 11 that the small-signal stability and stability margin of the multi-CBR system with SEs can be evaluated by the $gSCR$ and its critical value $CgSCR$. When $gSCR = 3.86 > CgSCR$, the 39-node system is stable since the convergence oscillation of active power outputs can be seen in Fig. 11(a). When $gSCR = 2.66 = CgSCR$, the 39-node system is critically stable since the undamped oscillation of active power outputs can be seen in Fig. 11(b). When $gSCR = 2.57 < CgSCR$,

TABLE IV
SENSITIVITY OF $\lambda_{\max}^+$ FOR S_{BEj} ON EACH NODE IN FIRST ITERATION

Node	Sen	Node	Sen	Node	Sen	Node	Sen
1	-0.0480	11	-0.0426	21	-0.0201	31	-0.0479
2	-0.0445	12	-0.0419	22	-0.0138	32	-0.0458
3	-0.0396	13	-0.0412	23	-0.0097	33	-0.0343
4	-0.0405	14	-0.0386	24	-0.0242	34	-0.0369
5	-0.0429	15	-0.0305	25	-0.0451	35	-0.0146
6	-0.0433	16	-0.0273	26	-0.0413	37	-0.0496
7	-0.0436	17	-0.0323	27	-0.0370	38	-0.0502
8	-0.0437	18	-0.0350	28	-0.0457	39	-0.0503
9	-0.0475	19	-0.0324	29	-0.0471		
10	-0.0422	20	-0.0343	30	-0.0479		

* "Sen" denotes the sensitivity of $\lambda_{\max}^+$ for S_{BEj} on each node.

TABLE V
GSCR, WEAKEST EIGENVALUES AND DAMPING RATIOS OF THE SYSTEM AFTER SUCCESSIVE PLACEMENT OF SEs BY PROPOSED METHOD

No. of installed SEs	$gSCR$	Weakest eigenvalue	Damping
0	2.650	$0.090 \pm 88.342i$	-0.001
1	2.926	$-1.758 \pm 88.949i$	0.020
2	3.256	$-3.485 \pm 89.346i$	0.039
3	3.666	$-5.104 \pm 89.564i$	0.057

the 39-node system is unstable since the divergent oscillation of active power outputs can be seen in Fig. 11(c). The time-domain simulation results verify the effectiveness of the small-signal stability assessment method based on $gSCR$ and $CgSCR$.

2) Verification of Proposed Optimal Placement Method:

a) Grid strength analysis and modal analysis: The effectiveness of the proposed SE's optimal placement method for enhancing the system's small-signal stability is verified on this 39-node system. Here, we consider that the system will be placed with three SEs. According to the proposed optimal placement method in Section V, we calculate the initial sensitivity of $\lambda_{\max}^+$ for SE_j 's capacity on each node as shown in Table IV. It can be seen from Table IV that node 39 is the optimal node for placing the first SE, since the corresponding sensitivity at node 39 is smallest. Similarly, the sensitivity of $\lambda_{\max}^+$ for SE_j 's capacity on each node after placing the first SE is given in Table XII. Table XII shows that node 38 is the best node for placing the second SE. Besides, Table XIII in Appendix II shows the sensitivity of $\lambda_{\max}^+$ for SE_j 's capacity on each node after placing two SEs. It can be seen from Table XIII that node 37 is the best node for placing the third SE. Therefore, the optimal placement of three SEs by the proposed method is nodes 37, 38 and 39.

The $gSCR$, weakest eigenvalues and corresponding damping ratio of the system when placing the SEs by the proposed optimal placement method are given in Table V. We can see from Table V that the $gSCR$ is increased from 2.650 to 3.666, and the damping ratio of the system's weakest eigenvalues is increased from -0.001 to 0.057 by the proposed method. Besides, to make a comparison, we consider two other cases: 1) the three SEs are all installed at node 23; 2) the three SEs are respectively installed at nodes 3, 22 and 35. Under these two cases, the system's weakest eigenvalues (or damping ratio) are $-0.976 \pm 88.714i$ and $-2.427 \pm 89.122i$ (or 0.011 and 0.027), respectively. In

TABLE VI
TOP EIGHTEEN LARGEST PARTICIPATION FACTORS AND CORRESPONDING
STATE VARIABLES FOR THE WEAKEST MODE

Variable	Par	Variable	Par	Variable	Par
θ_{pll9}	0.067	x_{i8}	0.064	θ_{pll5}	0.049
θ_{pll8}	0.067	x_{i7}	0.063	x_{i5}	0.047
θ_{pll7}	0.066	θ_{pll3}	0.061	θ_{pll4}	0.045
x_{i9}	0.064	x_{i1}	0.061	x_{i4}	0.043
θ_{pll1}	0.064	x_{i2}	0.061	θ_{pll6}	0.033
θ_{pll2}	0.064	x_{i3}	0.058	x_{i6}	0.026

* Par denotes participation; x_{ij} , ($j=1, \dots, 9$) denotes integral variable in PI of CBR i 's PLL; θ_{pll} denotes the angle of PLL.

comparison with these two cases, our proposed method can effectively improve the system's small-signal stability.

b) *Time-domain simulation*: Moreover, we provide the electromagnetic time-domain simulation using MATLAB/Simulink to verify the superiority of the proposed optimal placement method. Four cases are considered:

- Case 1): three SEs are uninstalled (as a reference);
- Case 2): three SEs are all installed at node 23;
- Case 3): three SEs are installed at nodes 3, 22 and 35;
- Case 4): three SEs are respectively installed at nodes 37~39.

In the above four cases, Case 4) represents the results by the proposed optimal placement method. When the coefficient m is set as 0.75, the corresponding gSCR for the above Cases 1)~4) are equal to 2.650, 2.801, 3.043 and 3.666, respectively. Under these conditions, a same disturbance is applied to the infinite bus, which causes the voltage drop by 10% at 0.10 s and then is cleared at 0.12 s. Fig. 12 shows the corresponding active power trajectories of all CBRs in the 39-node system following the disturbance.

It can be observed from Fig. 12 that the proposed optimal placement method can effectively improve the system's small-signal stability. As shown in Fig. 12(a)~(d), we can see that the convergence speed tends to be faster from Case 1) to Case 4), i.e., the system tends to be more stable. It demonstrates that the placement of SEs can improve the system's small-signal stability; besides, the placement of SEs by the proposed method improves the system's stability best. The time-domain simulation results verify the effectiveness of the proposed optimal placement method.

To demonstrate the robustness of the proposed method in Section V-A, we consider the variation of CBRs' control parameters. The above case with $m = 0.75$ is provided. Under this case, Table VI provides the top eighteen largest participation factors and corresponding state variables for the weakest mode. We can see from Table VI that the weakest mode (i.e., system's stability) is mainly related with CBRs' PLL. Besides, we note that the decrease of proportional parameter and increase of integral parameter in CBRs' PLL will deteriorate the system's stability, of which the same conclusions can refer to Refs. [15] and [16]. For clarity, we omit the detailed discussion about how the varying parameters in CBRs' PLL influence the system's stability.

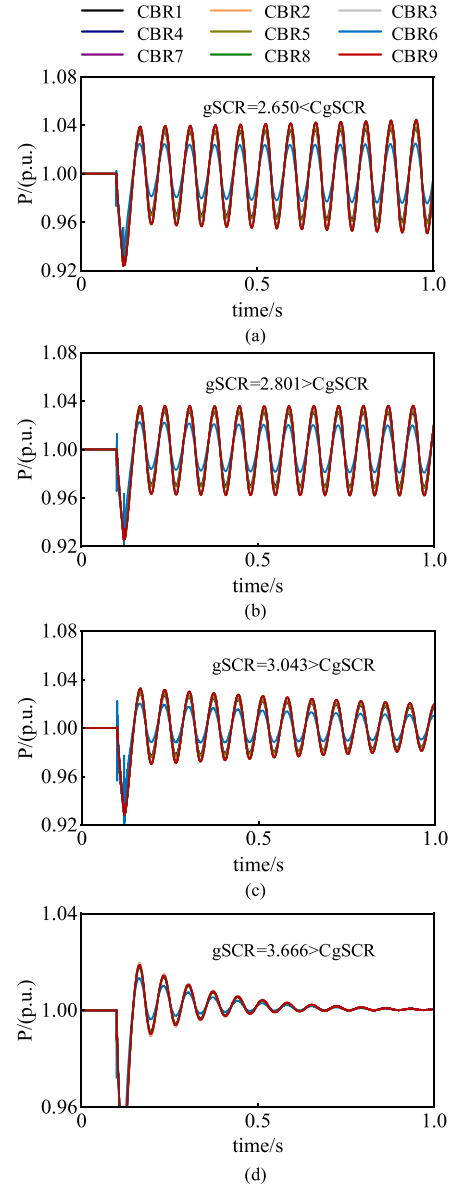


Fig. 12 Time-domain responses of active power outputs of all CBRs in the 39-node system under four cases: (a) Case 1); (b) Case 2); (c) Case 3); (d) Case 4).

We change the proportional and integral (PI) parameters in CBRs' PLL: PI parameters in CBRs 1~3 are set as "28, 7700", PI parameters in CBRs 7~9 are set as "24, 7900", and the remaining CBRs are unchanged. Table VII shows the damping ratio of the system's weakest eigenvalue after placing a SE at each node. We can see from Table VII that when placing the SE at node 39, the system's damping ratio is largest, i.e., the system is most stable. Therefore, node 39 is the optimal node for placing one SE. It is noteworthy that we traverse all possible cases and find that nodes 38 and 37 are the optimal node for placing second SE and third SE, respectively. For clarity, we omit providing the damping ratio of the system's weakest eigenvalue after placing second SE and third SE. The simulation results are consistent with the above results, when the CBRs are homogeneous. This demonstrates the

TABLE VII
DAMPING RATIO OF WEAKEST EIGENVALUE AFTER PLACING A SE AT NODE i

Node	Dam.	Node	Dam.	Node	Dam.	Node	Dam.
1	0.019	11	0.017	21	0.008	31	0.019
2	0.018	12	0.016	22	0.005	32	0.018
3	0.016	13	0.017	23	0.003	33	0.013
4	0.016	14	0.016	24	0.009	34	0.014
5	0.017	15	0.012	25	0.018	35	0.005
6	0.018	16	0.011	26	0.016	37	0.019
7	0.018	17	0.013	27	0.015	38	0.018
8	0.018	18	0.014	28	0.017	39	0.020
9	0.019	19	0.013	29	0.018		
10	0.017	20	0.013	30	0.019		

* Dam. denotes damping ratio of weakest eigenvalue.

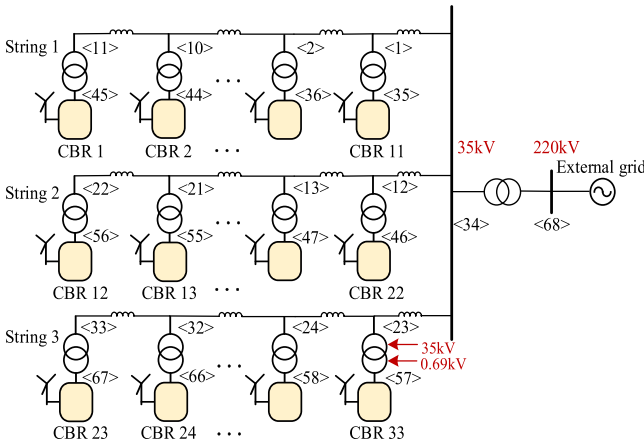


Fig. 13 A 33-converter system with a radian network topology.

validity and robustness of the proposed method for placing SEs, when considering slight variation of CBRs' control parameters.

C. Case Studies on a 33-Converter System

To demonstrate the validity of proposed method for placing SEs, A more complex power system with 33 converters is provided as shown in Fig. 13, where nodes 1~33 are connected to CBRs; CBRs connect the grid through a 0.69 kV/35 kV transformer and node 67 connects external grid through a 35 kV/ 220 kV transformer. Parameters of lines and transformers in the system refer to [20]. All CBRs' capacities are set as 1 p.u.. Here we consider to place 25% capacities of the CBRs in this system. That is, we place 11 SEs and each SE's capacity is set as 0.75 p.u.. The control parameters of all CBRs and SEs are presented in Table IX in Appendix II.

According to the proposed optimal placement method in Fig. 4, we calculate the initial sensitivity of $\lambda_{\max}^+$ for SE's capacity on each node as shown in Table VIII. It can be seen from Table VIII that node 67 is the optimal node for placing the first SE, since the corresponding sensitivity at node 67 is smallest. Similarly, the optimal placement for these 11 SEs are nodes 42, 43, 44, 45, 53, 54, 55, 56, 65, 66 and 67.

The electromagnetic time-domain simulation using MATLAB/Simulink is provided to verify the superiority of the proposed optimal placement method. Three cases are created:

Case 1): eleven SEs are uninstalled (as a reference);

TABLE VIII
SENSITIVITY OF $\lambda_{\max}^+$ FOR S_{BEj} ON EACH NODE IN FIRST ITERATION

Node	Sen	Node	Sen	Node	Sen	Node	Sen
1	-0.0017	18	-0.0023	35	-0.0083	52	-0.0114
2	-0.0018	19	-0.0024	36	-0.0090	53	-0.0117
3	-0.0019	20	-0.0024	37	-0.0096	54	-0.0120
4	-0.0020	21	-0.0025	38	-0.0101	55	-0.0121
5	-0.0021	22	-0.0025	39	-0.0106	56	-0.0122
6	-0.0022	23	-0.0017	40	-0.0111	57	-0.0083
7	-0.0023	24	-0.0018	41	-0.0114	58	-0.0090
8	-0.0024	25	-0.0019	42	-0.0117	59	-0.0096
9	-0.0024	26	-0.0020	43	-0.0120	60	-0.0101
10	-0.0025	27	-0.0021	44	-0.0121	61	-0.0106
11	-0.0025	28	-0.0022	45	-0.0122	62	-0.0111
12	-0.0017	29	-0.0023	46	-0.0083	63	-0.0114
13	-0.0018	30	-0.0024	47	-0.0090	64	-0.0117
14	-0.0019	31	-0.0024	48	-0.0096	65	-0.0120
15	-0.0020	32	-0.0025	49	-0.0101	66	-0.0121
16	-0.0021	33	-0.0025	50	-0.0106	67	-0.0122
17	-0.0022	34	-0.0015	51	-0.0111		

* "Sen" denotes the sensitivity of $\lambda_{\max}^+$ for S_{BEj} on each node.

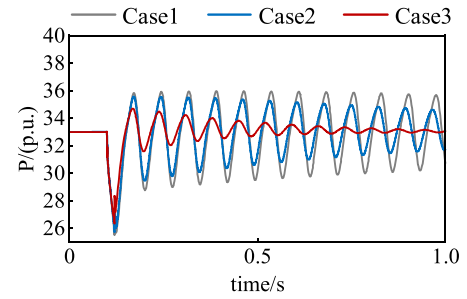


Fig. 14. Time-domain responses of active power outputs of all CBRs in the 33-converter system under three cases: (a) Case 1); (b) Case 2); (c) Case 3).

Case 2): eleven SEs are all installed at nodes 29~39;

Case 3): eleven SEs are respectively installed at nodes 42, 43, 44, 45, 53, 54, 55, 56, 65, 66 and 67.

In the above three cases, Case 3) represents the results by the proposed method; besides, we equally proportionally increase the network lines' reactance with coefficient m and $m = 4$, which is proportional to the line length in the system. The corresponding $gSCR = 2.823, 3.059$ and 3.308 for the above Cases 1)~3). Under these conditions, a disturbance is applied to the infinite bus. The disturbance causes the voltage to drop by 25% at 0.10s and then is cleared at 0.12s. Fig. 14 shows the corresponding active power trajectories of all CBRs in the 33-converter system following the disturbance.

We can see from Fig. 14 that the convergence speed of the system become faster from Case 1) to Case 3), which means the system becomes more stable. This demonstrates that the placement of SEs can improve the system's small-signal stability. Moreover, the system is more stable under Case 3) than that under Case 2), which verifies the effectiveness of the proposed optimal placement method.

VII. CONCLUSION AND DISCUSSIONS

This paper presented a method for optimally placing SEs to improve the small-signal stability of a power system integrated

with CBRs. By creating the frequency-domain impedance model for a multi-CBR system with SEs for the small-signal stability analysis, we revealed the theoretical relationship between SEs and the PLL-induced small-signal stability from the respective of grid strength. Particularly, it is found that when SE absorbs active power from the power grid, they can enhance grid strength and thus the small-signal stability. Moreover, the degree of system stability improvement depends on the location of SE placement. As a result, we proposed a grid strength-based method for optimally placing SEs to enhance the system small-signal stability. This proposed method is computationally efficient for the large multi-CBR system, even when the CBRs have “black-boxed” modeling. The efficacy of the proposed method was verified on three test systems.

The analysis results presented in this paper are based on the assumption that SEs have the similar control configuration and control parameters as CBRs. When these assumptions are not satisfied, the analysis results may be invalid. That is, the impact of SEs (when absorbing active power from the grid) on the PLL-induced small-signal stability may be different, if SEs have totally different control configuration or control parameters compared with CBRs. In the future work, we will use the matrix perturbation theory and robust analysis to explore how the SEs with heterogenous control configuration and control parameters influence the PLL-induced small-signal stability and how to place SEs in a multi-CBR system for enhancing the stability.

APPENDIX I

A. Proof of Proposition 1

Proof: 1) Since matrix $\mathbf{B}_{\text{red}}$ is Hermitian, strictly diagonally dominant and its diagonal elements are all positive, $\mathbf{B}_{\text{red}}$ is positive definite (Theorem 7.2.3 in [24]). Besides, since $\mathbf{B}_{\text{red}}$ is real symmetric, $\mathbf{B}_{\text{red}}$ can be written as $\mathbf{B}_{\text{red}} = \mathbf{Q}^T \text{diag}\{\lambda_i\} \mathbf{Q}$, ($i = 1, \dots, n$) where $\mathbf{Q} \in \mathbb{R}^{(n+k) \times (n+k)}$ is real orthogonal and λ_i are real positive eigenvalues of $\mathbf{B}_{\text{red}}$ (Theorem 4.1.5 in [24]). Therefore, $\mathbf{B}_{\text{red}}^{1/2}$ can be written as $\mathbf{B}_{\text{red}}^{1/2} = \mathbf{Q}^T \text{diag}\{\lambda_i^{1/2}\} \mathbf{Q}$, which demonstrates that $\mathbf{B}_{\text{red}}^{1/2}$ is real positive definite.

Then, we obtain that $\mathbf{B}_{\text{red}}^{1/2} (\mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}) \mathbf{B}_{\text{red}}^{-1/2} = \mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$, which demonstrates that $\mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}$ is similar to $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$. $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$ is real symmetric (or Hermitian), which demonstrates that eigenvalues of $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$ (or $\mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}$) are all real values (Corollary 2.5.11 in [24]);

2) Since $\mathbf{B}_{\text{red}}^{1/2}$ is nonsingular, $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$ is congruent to $\mathbf{S}_{\text{B1}}^{-1}$. Therefore, $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$ and $\mathbf{S}_{\text{B1}}^{-1}$ have the same inertia (Theorem 4.5.8 in [24]). Besides, $\mathbf{S}_{\text{B1}}^{-1}$ is diagonal with n real positive diagonal elements and k real negative diagonal elements, which demonstrates that $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$ has n real positive eigenvalues and k real negative eigenvalues. Besides, since $\mathbf{B}_{\text{red}}^{1/2} \mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}^{1/2}$ is similar to $\mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}$, the conclusion is proved.

B. Proof of Proposition 2

Proof: Before proving this proposition, we demonstrate that it satisfies $\lambda_{\min}^+ (\mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}) = 1/\lambda_{\max}^+ (\mathbf{S}'_{\text{B1}} \mathbf{Z})$. In other words,

the minimal positive eigenvalue of $\mathbf{S}_{\text{B1}}^{-1} \mathbf{B}_{\text{red}}$ (i.e., gSCR) is the reciprocal of the maximal positive eigenvalue of $\mathbf{S}'_{\text{B1}} \mathbf{Z}$, i.e.,

$$\text{gSCR} = 1/\lambda_{\max}^+ (\mathbf{S}'_{\text{B1}} \mathbf{Z}), \mathbf{S}'_{\text{B1}} = \begin{bmatrix} \mathbf{S}_{\text{B1}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (19)$$

where $\mathbf{Z} \in \mathbb{R}^{(n+m) \times (n+m)}$ is network impedance matrix (containing nodes $1 \sim n+m$), and $\lambda_{\max}^+()$ is maximal eigenvalue of a matrix, $\mathbf{0}$ represents a zero square matrix, $\mathbf{S}'_{\text{B1}} \in \mathbb{R}^{(n+m) \times (n+m)}$.

The proof of (19) refers to [25]. Note that (19) can consider the case that some of diagonal elements of $\mathbf{S}'_{\text{B1}}$ are zero, (i.e., corresponding nodes are not placed with CBRs or SEs).

Then, to prove this proposition, we consider the scenario that only one SE is connected to the multi-CBR system. When not considering the connection of SEs, $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ is rewritten as $\begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} \mathbf{Z}$. Similar as the derivation in the above proof of Proposition 1, it can be concluded that $\begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} \mathbf{Z}$ is similar to $\mathbf{Z}^{1/2} \begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} \mathbf{Z}^{1/2}$ and congruent to $\begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix}$. Therefore,

$\begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} \mathbf{Z}$ has n positive eigenvalues and k zero eigenvalues, of which the eigenvalues can be written as $0, \dots, 0 < \lambda'_1 \leq \dots \leq \lambda'_n$. When the multi-CBR system is applied with a SE at node j , the weighted admittance matrix $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ is written as $(\begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} - S_{\text{BE}j} \mathbf{E}_j) \mathbf{Z}$, ($\mathbf{E}_j$ is a square matrix, where only the j th diagonal element is 1 and the other elements are zero).

Define the maximal eigenvalues of the corresponding $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ as $\bar{\lambda}'_n$. We note that $(\begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} - S_{\text{BE}j} \mathbf{E}_j) \mathbf{Z}$ is similar to

$\mathbf{Z}^{1/2} \begin{bmatrix} \mathbf{S}_{\text{Bn}} & \mathbf{0} \end{bmatrix} \mathbf{Z}^{1/2} - S_{\text{BE}j} \mathbf{v}'_j \mathbf{v}'_j^T$, where $\mathbf{v}'_j$ is the j th column vector of $\mathbf{Z}^{1/2}$ and $S_{\text{BE}j} > 0$. Therefore, according to the corollary 4.3.9 in [24], the maximal eigenvalue of $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ under two conditions whether considering the connection of the SE or not satisfies the following inequalities:

$$0 < \bar{\lambda}'_n \leq \lambda'_n \Rightarrow 0 < \frac{1}{\lambda'_n} \leq \frac{1}{\bar{\lambda}'_n} \quad (20)$$

where λ'_n is the maximal eigenvalue of $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ when not considering the connection of a SE; $\bar{\lambda}'_n$ is the maximal eigenvalue of $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ when considering the connection of a SE.

According to (20), it is concluded that the connection of a SE (when absorbing active powers from the grid) increases gSCR. When connecting more SEs in the multi-CBR system, we can conclude from iterative idea that the connection of SEs (when absorbing active powers) always increases the gSCR.

C. Derivation of (16)

Define $\mathbf{v}_1$ and $\mathbf{w}_1$ as the left and right eigenvectors of $\mathbf{S}'_{\text{B1}} \mathbf{Z}$ for the maximal eigenvalue $\lambda_{\max}^+$, which satisfies that

$$\begin{cases} \mathbf{v}_1 \mathbf{S}'_{\text{B1}} \mathbf{Z} = \lambda_{\max}^+ \mathbf{v}_1 \\ \mathbf{S}'_{\text{B1}} \mathbf{Z} \mathbf{w}_1 = \lambda_{\max}^+ \mathbf{w}_1 \\ \mathbf{v}_1 \mathbf{w}_1 = 1 \end{cases} \quad (21)$$

The first equation in (21) can be rewritten as

$$\mathbf{Z} \mathbf{S}'_{\text{B1}} \mathbf{v}_1^T = \lambda_{\max}^+ \mathbf{v}_1^T \Rightarrow \mathbf{S}'_{\text{B1}} \mathbf{Z} \mathbf{S}'_{\text{B1}} \mathbf{v}_1^T = \lambda_{\max}^+ \mathbf{S}'_{\text{B1}} \mathbf{v}_1^T \quad (22)$$

It can be concluded from (21) and (22) that

$$\mathbf{w}_1 = a\mathbf{S}'_{B1}\mathbf{v}_1^T \quad (23)$$

By taking the derivative of $\lambda_{\max}^+$ for S_{BEj} in the second equation of (21), we obtain that:

$$-\mathbf{E}_j\mathbf{Z}\mathbf{w}_1 + \mathbf{S}'_{B1}\mathbf{Z}\frac{\partial\mathbf{w}_1}{\partial S_{BEj}} = \frac{\partial\lambda_{\max}^+}{\partial S_{BEj}}\mathbf{w}_1 + \lambda_{\max}^+\frac{\partial\mathbf{w}_1}{\partial S_{BEj}} \quad (24)$$

Left-multiplying $\mathbf{v}_1$ in (24), we obtain that

$$-\mathbf{v}_1\mathbf{E}_j\mathbf{Z}\mathbf{w}_1 = \frac{\partial\lambda_{\max}^+}{\partial S_{BEj}}\mathbf{v}_1\mathbf{w}_1 = \frac{\partial\lambda_{\max}^+}{\partial S_{BEj}} \quad (25)$$

Substituting (23) into (25) obtains that

$$\begin{aligned} \frac{\partial\lambda_{\max}^+}{\partial S_{BEj}} &= -a\mathbf{v}_1\mathbf{E}_j\mathbf{Z}\mathbf{S}'_{B1}\mathbf{v}_1^T \Leftrightarrow \frac{\partial\lambda_{\max}^+}{\partial S_{BEj}} \\ &= \lambda_{\max}^+a\mathbf{v}_1\mathbf{E}_j\mathbf{v}_1^T = -a\lambda_{\max}^+v_{j1}^2 \end{aligned} \quad (26)$$

As discussed in Section IV-B, the placement of SEs increases gSCR (or decrease $\lambda_{\max}^+$). Therefore, a is a positive real number. Equation (26) shows that the sensitivity of $\lambda_{\max}^+$ for S_{BEj} on each node is always smaller (or equal) to zero (i.e., increase gSCR), which is consistent with the conclusion in Proposition 2.

APPENDIX II

TABLE IX

CONTROL PARAMETERS OF CBRs AND SES (PER-UNIT VALUES)

Filter inductance L_f , filter capacitance C_f , dc capacitance C_{dc} : 0.05, 0.05, 0.038
Transfer function of the current control loop $H_i(s)$: $1+10/s$
Transfer function of the active power control loop $H_p(s)$: $0.5+5/s$
Transfer function of the voltage feedforward filter $G_{FF}(s)$: $1/(1+0.01s)$
Transfer function of the PLL $H_{pll}(s)$: $(26+7800s)/s$
Parameters of active power output reference and q -axis current reference: 1, 0 for CBRs (or -1, 0 for SEs)

TABLE X

LINE PARAMETERS AND CBR CAPACITIES IN TWO-AREA FOUR-CBR SYSTEM

$X_{1,5}=0.152$	$X_{2,5}=0.049$	$X_{3,9}=0.039$	$X_{4,9}=0.201$
$X_{5,6}=0.039$	$X_{6,7}=0.029$	$X_{7,8}=0.017$	$X_{8,9}=0.039$
$S_{B1}=0.5$	$S_{B2}=1$	$S_{B3}=1.5$	$S_{B4}=0.5$

TABLE XI

NETWORK LINES' REACTANCE IN THE IEEE 39-BUS SYSTEM

Line _{1,2}	0.0411	Line _{1,39}	0.0250	Line _{2,3}	0.0151
Line _{2,30}	0.0181	Line _{3,4}	0.0213	Line _{3,18}	0.0133
Line _{4,14}	0.0129	Line _{5,8}	0.0112	Line _{5,6}	0.0026
Line _{6,11}	0.0082	Line _{7,8}	0.0046	Line _{8,9}	0.0363
Line _{10,11}	0.0043	Line _{10,13}	0.0043	Line _{10,32}	0.0200
Line _{12,13}	0.0435	Line _{13,14}	0.0101	Line _{14,15}	0.0217
Line _{16,17}	0.0089	Line _{16,19}	0.0195	Line _{16,21}	0.0135
Line _{17,18}	0.0082	Line _{17,27}	0.0173	Line _{19,33}	0.0142
Line _{20,34}	0.0180	Line _{21,22}	0.0140	Line _{22,23}	0.0096
Line _{23,24}	0.0350	Line _{23,36}	0.0272	Line _{25,26}	0.0323
Line _{26,27}	0.0147	Line _{26,28}	0.0474	Line _{26,29}	0.0625
Line _{29,38}	0.0156	Line _{6,31}	0.0250	Line _{19,20}	0.0138
Line _{2,25}	0.0086	Line _{11,12}	0.0435	Line _{22,35}	0.0143
Line _{4,5}	0.0128	Line _{15,16}	0.0094	Line _{25,37}	0.0232
Line _{6,7}	0.0092	Line _{16,24}	0.0059	Line _{28,29}	0.0151
Line _{9,39}	0.0250	x_T	0.0001		

TABLE XII

SENSITIVITY OF $\lambda_{\max}^+$ FOR S_{BEj} ON EACH NODE IN SECOND ITERATION

Node	Sen	Node	Sen	Node	Sen	Node	Sen
1	-0.0443	11	-0.0419	21	-0.0203	31	-0.0475
2	-0.0438	12	-0.0413	22	-0.0140	32	-0.0456
3	-0.0392	13	-0.0407	23	-0.0098	33	-0.0354
4	-0.0399	14	-0.0382	24	-0.0244	34	-0.0384
5	-0.0421	15	-0.0306	25	-0.0447	35	-0.0149
6	-0.0424	16	-0.0275	26	-0.0415	37	-0.0497
7	-0.0424	17	-0.0324	27	-0.0372	38	-0.0515
8	-0.0424	18	-0.0349	28	-0.0464	39	-0.0447
9	-0.0437	19	-0.0333	29	-0.0481		
10	-0.0417	20	-0.0355	30	-0.0475		

TABLE XIII

SENSITIVITY OF $\lambda_{\max}^+$ FOR S_{BEj} ON EACH NODE IN THIRD ITERATION

Node	Sen	Node	Sen	Node	Sen	Node	Sen
1	-0.0445	11	-0.0429	21	-0.0205	31	-0.0492
2	-0.0435	12	-0.0422	22	-0.0142	32	-0.0472
3	-0.0391	13	-0.0416	23	-0.0099	33	-0.0368
4	-0.0405	14	-0.0388	24	-0.0246	34	-0.0403
5	-0.0429	15	-0.0309	25	-0.0440	35	-0.0152
6	-0.0433	16	-0.0277	26	-0.0385	37	-0.0494
7	-0.0433	17	-0.0320	27	-0.0355	38	-0.0408
8	-0.0433	18	-0.0347	28	-0.0397	39	-0.0451
9	-0.0444	19	-0.0344	29	-0.0401		
10	-0.0427	20	-0.0368	30	-0.0476		

REFERENCES

- [1] F. Blaabjerg, Y. Yang, K. A. Kim, and J. Rodriguez, "Power electronics technology for large-scale renewable energy generation," *Proc. IEEE*, vol. 111, no. 4, pp. 335–355, Apr. 2023.
- [2] Y. Gu and T. C. Green, "Power system stability with a high penetration of inverter-based resources," *Proc. IEEE*, vol. 111, no. 7, pp. 832–853, Jul. 2023.
- [3] X. Li, D. Hui, and X. Lai, "Battery energy storage station (BESS)-based smoothing control of photovoltaic (PV) and wind power generation fluctuations," *IEEE Trans. Sustain. Energy*, vol. 4, no. 2, pp. 464–473, Apr. 2013.
- [4] Z. Zhao, H. Hu, Z. He, H. Zhu, P. Davari, and F. Blaabjerg, "Advanced solid-state lithium battery and its safety," *CPSS Trans. Power Electron. Appl.*, vol. 8, no. 4, pp. 348–362, Dec. 2023.
- [5] F. Z. Peng, C. C. Liu, Y. Li, A. K. Jain, and D. Vinnikov, "Envisioning the future renewable and resilient energy grids—A power grid revolution enabled by renewables, energy storage, and energy electronics," *IEEE J. Emerg. Sel. Top. Ind. Electron.*, vol. 5, no. 1, pp. 8–26, Jan. 2024.
- [6] P. Roy, J. He, T. Zhao, and Y. V. Singh, "Recent advances of wind-solar hybrid renewable energy systems for power generation: A review," *IEEE Open J. Ind. Electron. Soc.*, vol. 3, pp. 81–104, 2022.
- [7] F. Calero et al., "A review of modeling and applications of energy storage systems in power grids," *Proc. IEEE*, vol. 111, no. 7, pp. 806–831, Jul. 2023.
- [8] B. Fang, X. Ren, W. Lin, H. Li, G. He, and W. Zhou, "Analysis of the influence of large-scale integration of centralized energy storage into the power grid on voltage security and stability of power system," in *Proc. IEEE 3rd Asia Energy Elect. Eng. Symp.*, 2021, pp. 759–764.
- [9] M. A. K. Magableh, A. A. A. Radwan, Y. A.-R. I. Mohamed, and E. F. El-Saadany, "A novel reduced-order modeling approach of a grid-tied hybrid photovoltaic–wind turbine–battery energy storage system for dynamic stability analysis," *IEEE Open J. Power Electron.*, vol. 5, pp. 1459–1483, 2024.
- [10] M. Ghofrani, A. Arabali, M. Etezadi-Amoli, and M. S. Fadali, "A framework for optimal placement of energy storage units within a power system with high wind penetration," *IEEE Trans. Sustain. Energy*, vol. 4, no. 2, pp. 434–442, Apr. 2013.
- [11] J. Zhang, N. Zhang, and Y. Ge, "Energy storage placements for renewable energy fluctuations: A practical study," *IEEE Trans. Power Syst.*, vol. 38, no. 5, pp. 4916–4927, Sep. 2023.

- [12] Q. Sun, B. Huang, D. Li, D. Ma, and Y. Zhang, "Optimal placement of energy storage devices in microgrids via structure preserving energy function," *IEEE Trans. Ind. Informat.*, vol. 12, no. 3, pp. 1166–1179, Jun. 2016.
- [13] R. Liu, J. Yao, X. Wang, P. Sun, J. Pei, and J. Hu, "Dynamic stability analysis and improved LVRT schemes of DFIG-based wind turbines during a symmetrical fault in a weak grid," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 303–318, Jan. 2020.
- [14] L. Fan, "Modeling type-4 wind in weak grids," *IEEE Trans. Sustain. Energy*, vol. 10, no. 2, pp. 853–864, Apr. 2019.
- [15] J. Z. Zhou, H. Ding, S. Fan, Y. Zhang, and A. M. Gole, "Impact of short-circuit ratio and phase-locked-loop parameters on the small-signal behavior of a VSC-HVDC converter," *IEEE Trans. Power Del.*, vol. 29, no. 5, pp. 2287–2296, Oct. 2014.
- [16] L. Xu, H. Xin, L. Huang, H. Yuan, P. Ju, and D. Wu, "Symmetric admittance modeling for stability analysis of grid-connected converters," *IEEE Trans. Energy Convers.*, vol. 35, no. 1, pp. 434–444, Mar. 2020.
- [17] Á. Ortega and F. Milano, "Generalized model of VSC-based energy storage systems for transient stability analysis," *IEEE Trans. Power Syst.*, vol. 31, no. 5, pp. 3369–3380, Sep. 2016.
- [18] W. Dong, H. Xin, D. Wu, and L. Huang, "Small-signal stability analysis of multi-infeed power electronic systems based on grid strength assessment," *IEEE Trans. Power Syst.*, vol. 34, no. 2, pp. 1393–1403, Mar. 2019.
- [19] Y. Hao, C. Chen, G. Lu, L. Qu, X. Wang, and Y. Zhou, "Identification method of weak links based on grid strength assessment," in *Proc. IEEE 6th Int. Elect. Energy Conf.*, 2023, pp. 2548–2552.
- [20] Y. Zhou et al., "Small-signal stability assessment of heterogeneous grid-following converter power systems based on grid strength analysis," *IEEE Trans. Power Syst.*, vol. 38, no. 3, pp. 2566–2579, May 2023.
- [21] H. Yuan, H. Xin, D. Wu, W. Wang, and Y. Zhou, "Small-signal stability assessment of multi-converter-based-renewable systems with STATCOMs based on generalized short-circuit ratio," *IEEE Trans. Energy Convers.*, vol. 37, no. 4, pp. 2889–2902, Dec. 2022.
- [22] Y. Zhou et al., "Small-signal stability analysis of power systems under uncertain renewable generation," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2022, pp. 1–5.
- [23] C. Liu, H. Xin, D. Wu, H. Gao, H. Yuan, and Y. Zhou, "Generalized operational short-circuit ratio for grid strength assessment in power systems with high renewable penetration," *IEEE Trans. Power Syst.*, vol. 39, no. 4, pp. 5479–5494, Jul. 2024.
- [24] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge Univ. Press: New York, NY, USA, 2013.
- [25] H. Yuan, "Assessing maximal capacity of grid-following converters with grid strength constraints," *IEEE Trans. Sustain. Energy*, vol. 13, no. 4, pp. 2119–2132, Oct. 2022.



power system stability analysis and control.

Hui Yuan received the B.Eng. degree from the College of Automation, Wuhan University of Technology, Wuhan, China, in 2013, the M.Eng. degree from the College of Electrical Engineering, Guangxi University, Nanning, China, in 2016, and the Ph.D. degree from the College of Electrical Engineering, Zhejiang University, Hangzhou, China, in 2021. From 2021 to 2024, he was a Postdoctor with Zhejiang University. He is currently an Associated Professor with the College of Electrical Engineering, Guangxi University. His research interests include renewable



Yi Hao received the B.Eng. degree from the College of Electrical Engineering, Shandong University, Jinan, China, in 2021, and the M.Eng. degree from the College of Electrical Engineering, Zhejiang University, Hangzhou, China, in 2024. She is currently an Engineer with State Grid Shandong Electric Power Research Institute. Her research interests include power dispatching operation analysis and control.



Huanhai Xin (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical Engineering, Zhejiang University, Hangzhou, China, in 2007. From 2009 to 2010, he was a Postdoctoral Researcher with the Department of Electrical Engineering and Computer Science, University of Central Florida, Orlando, FL, USA. He is currently a Professor with the College of Electrical Engineering, Zhejiang University. He is currently supported by the National Natural Science of China for Excellent Young Scholars. His research interests include renewable power system stability analysis and control.



Linbin Huang (Member, IEEE) received the B.Eng. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 2015 and 2020, respectively. He is currently a ZJU100 Young Professor with Zhejiang University, Hangzhou, China. From 2020 to 2024, he was with ETH Zürich, Switzerland, where he was a Postdoctoral Researcher and then promoted to a Senior Scientist. His research interests include power system stability, optimal control of power electronics, and data-driven control.



Yuhao Zhou received the B.Eng. degree in electrical and electronic engineering, North China Electric Power University, Beijing, China, in 2019, and the Ph.D. degree from the College of Electrical Engineering, Zhejiang University, Hangzhou, China, in 2024. He is currently with the China Electric Power Planning Engineering Institute. His research interests include renewable power system stability analysis and control.



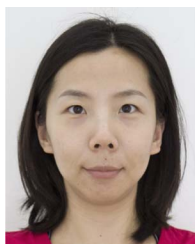
Xiaofei Wang received the B.Eng. degree from the Department of Electrical and Electronic Engineering, University of Hong Kong, Hong Kong, in 2012, and the M.A.Sc. degree from the Edward S. Rogers Sr. Department of Electrical & Computer Engineering, University of Toronto, Toronto, ON, Canada, in 2016. She is currently a Laboratory Technician with the College of Electrical Engineering, Zhejiang University, Hangzhou, China. Her research interests include stability and control of power systems with high renewable penetration.



Chunmeng Chen received the master's degree in engineering. He has been engaged with the Technical Management of Provincial Power Grid Dispatching for more than 15 years. He has extensive experience in the operation, stability analysis, and transaction balancing of power system. He has participated in the power system stability analysis of major projects such as Qinghai Tibet Interconnection, Yushu Interconnection, and Qingyu Ultra-HVDC, and formulated power grid operation control principles.



Guoqiang Lu received the B.Eng. degree from the Department of Electrical and Electronic Engineering, Lanzhou Jiaotong University, Lanzhou, China. He is currently the Deputy Director of the System Operation Department of State Grid Qinghai Electric Power Dispatching and Control Center. His research interests include power system analysis and operation control, power and energy balance analysis, new energy grid research and management, major project commissioning and debugging, new energy consumption analysis and improvement research.



Linan Qu received the master's degree in electrical engineering from the Harbin Institute of Technology, Harbin, China, in 2010 and the doctoral degree in electrical engineering from Hebei University of Technology, Tianjin, China, in 2024. She is currently with the Science and Technology Research Institute of China Electrical Equipment Group. She is also the Head of the Renewable Energy Power Generation Specialty. Her research interests include simulation analysis and operation control of renewable energy generation connected to power systems.



Di Wu (Member, IEEE) received the B.S. degree in electrical engineering from Zhejiang University, Hangzhou, China, in 2002, the M.S. degree in electrical engineering from the Kunming University of Science and Technology, Kunming, China, in 2005, the Ph.D. degree in the area of applied complex networks from the Polytechnic University of Turin, Turin, Italy, in 2011 and the second Ph.D. degree in the area of power system engineering from Zhejiang University, Hangzhou, China. He is currently an Assistant Professor with the Department of Electrical and Computer Engineering, North Dakota State University, Fargo, ND, USA. His research interests include renewable energy integration, power system dynamics and stability, cascading failures, and application of complex network theory in power grids.